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Test Plan

A comprehensive test plan covers normal operation, failure cases, and malicious inputs. Each test has a unique ID, purpose, steps, conditions, expected result, and type (E = Expected, F = Failure, M = Malicious).

1. Temperature Reading Object Tests

ID	Purpose	Steps	Conditions	Expected Result	Type
TRE1	Create Celsius object	`TemperatureReading(20, "Celsius")`	Valid value and unit	Object created correctly	E
TRE2	Create Fahrenheit object	`TemperatureReading(77, "Fahrenheit")`	Valid value and unit	Object created correctly	E
TRE3	Create Kelvin object	`TemperatureReading(300, "Kelvin")`	Valid value and unit	Object created correctly	E
TRF1	Empty unit	`TemperatureReading(20, "")`	Unit is empty	Program shows ValueError	F
TRF2	Wrong unit name	`TemperatureReading(20, "Temperature")`	Invalid unit	Program shows ValueError	F

ID	Purpose	Steps	Conditions	Expected Result	Type
TRF3	Invalid Kelvin value	`TemperatureReading(-10, "Kelvin")`	Below absolute zero	Program shows ValueError	F
TRF4	Wrong value type	`TemperatureReading("Celsius", 20)`	Value is not a number	Program shows error	F
TR-M1	Malicious input	`TemperatureReading(20, "import('os')")`	Bad input (code injection attempt)	Program shows error, stays safe	M

2. Temperature Conversion Tests

ID	Purpose	Steps	Conditions	Expected Result	Type
TC-E1	Celsius to Celsius	Call `Celsius()` on Celsius object	Normal input	Returns correct value (20)	E
TC-E2	Celsius to Fahrenheit	Call `Fahrenheit()` on 20°C object	Normal input	Returns 68	E
TC-E3	Celsius to Kelvin	Call `Kelvin()` on 0°C object	Normal input	Returns 273.15	E
TC-E4	Fahrenheit to Celsius	Call `Celsius()` on 32°F object	Normal input	Returns 0	E
TC-E5	Kelvin to Celsius	Call `Celsius()` on 273.15K object	Normal input	Returns 0	E
TC-F1	Invalid value	Call method on object with `None` value	Bad object	Program shows error	F
TC-F2	Very large number	Call conversion on extremely large value	Edge case	Program handles safely or shows error	F

TC-M1	Malicious value	Call conversion with value set to `"import()"`	Bad input	Program shows error, stays safe	M

3. Temperature Analysis Tests

ID	Purpose	Steps	Conditions	Expected Result	Type
TA-E1	Filter readings	Call `filter_by_time()` with valid timestamps	Valid data	Returns correct readings	E
TA-E2	Find minimum	Call `minimum()` with valid data	Valid data	Returns correct minimum	E
TA-E3	Find maximum	Call `maximum()` with valid data	Valid data	Returns correct maximum	E
TA-E4	Find mean	Call `mean()` with valid data	Valid data	Returns correct mean	E

TA-E5	Find standard deviation	Call `std_dev()` with valid data	Valid data	Returns correct result	E
TA-F1	Wrong time range	Start time after end time	Invalid input	Program shows error	F
TA-F2	No readings in range	Call analysis with empty filtered list	No data in range	Program shows error	F
ID	Purpose	Steps	Conditions	Expected Result	Type
TA-F3	Empty data list	Create `TemperatureAnalysis` with empty lists	Missing data	Program shows error	F
TA-M1	Malicious input	Use `"import()"` as timestamp	Bad input	Program shows error, stays safe	M